

Fused determiner-heads

Fragment 1 — 2026-05-07

This fragment tests the local end of the project: can a CGEL-style category / function distinction be stated as a small constraint on labelled trees? The target is the familiar contrast between an NP with separate daughters in Determiner and Head functions, and an NP in which one determiner realizes both functions.

1. Phenomenon

CGEL treats examples such as *three*, *some*, and *many* in independent NP use as cases of fused determiner-heads. The fusion is stated over functions: the same daughter fills Determiner and Head. The fused determiner is a DP in the ordinary simple cases; in genitive cases it can be a genitive NP.

- (1) *Pat invited two students; I invited three.*
- (2) *Some of the reports were late, but many arrived on time.*
- (3) *Kim's proposal was stronger than Pat's.*

The descriptive problem is small but useful. In *two students*, *two* is the determiner, and the nominal *students* fills Head. In independent *three*, the determiner fills both functions. In *Pat's*, the determiner is a genitive NP. A fragment should therefore distinguish ordinary determiner-plus-Head NPs from fused determiner-head NPs without recasting the fused function pair as either a lexical category claim or a semantic category claim.

2. Vocabulary

Use the shared relations $\triangleleft$ and $\triangleleft^*$, and the function relations Head and Det. Add one local syntactic predicate for determiners in this draft:

- (4) $\text{SynDeterminer}(x) \equiv_{\text{def}} \text{DP}(x) \vee \text{GenNP}(x)$
x is a syntactic determiner: a DP or a genitive NP in this draft

The predicate GenNP is intentionally provisional: a fuller fragment should say whether it is a feature-bearing subtype of NP, a separate abbreviation, or a definable configuration. For now it keeps the category/function issue visible.

No semantic predicate is introduced here. In particular, $\text{SynDeterminer}(x)$ does not say that *x* denotes a generalized quantifier, a choice function, a possessive relation, or any other semantic category. If a later fragment needs that bridge, it should add a separate interpretation relation, such as $\text{InterpretsAs}(x, s)$, with explicit warrants.

3. Specification

First define the fused function-pair relation:

- (5) $\text{FusedDetHead}(d, n) \equiv_{\text{def}} n \triangleleft d \wedge \text{Det}(n, d) \wedge \text{Head}(n, d)$.
d is both Determiner and Head of n

An ordinary NP with separate Determiner and nominal Head daughters can be characterised as follows:

- (6) $\text{CanonicalDetHeadNP}(n) \equiv_{\text{def}} \text{NP}(n) \wedge (\exists d)(\exists h)[n \triangleleft d \wedge n \triangleleft h \wedge d \neq h$
 $\wedge \text{Det}(n, d) \wedge \text{SynDeterminer}(d) \wedge \text{Head}(n, h) \wedge \text{Nom}(h)]$.
an NP with a determiner and a distinct nominal Head

The fused determiner-head configuration is the corresponding case where one determiner realizes both functions and no separate nominal Head daughter is present:

- (7) $\text{FusedDetHeadNP}(n) \equiv_{\text{def}} \text{NP}(n) \wedge (\exists d)[\text{SynDeterminer}(d) \wedge \text{FusedDetHead}(d, n)$
 $\wedge \neg(\exists h)[n \triangleleft h \wedge h \neq d \wedge \text{Head}(n, h) \wedge \text{Nom}(h)]]$.
an NP whose determiner is both Determiner and Head

For simple cases such as *three*, this is a first-order local definition. Partitive cases add one more daughter, but the fused determiner is still the DP in the fused function pair:

- (8) $\text{PartitiveFusedDetHeadNP}(n) \equiv_{\text{def}} \text{FusedDetHeadNP}(n) \wedge (\exists p)[n \triangleleft p \wedge \text{Comp}(n, p) \wedge \text{PP}(p)]$.
a fused determiner-head NP with an of-PP complement

Nothing in the formula assigns the fused element to a semantic category. It says only that this tree has one daughter bearing two syntactic functions. In the ordinary cases that daughter is a DP.

4. Worked instance

Take *I invited three*. Let n be the object NP and d the DP *three*. The intended tree makes d a determiner and gives d both the Determiner and Head functions in n . There is no distinct nominal Head daughter, so n satisfies $\text{FusedDetHeadNP}(n)$.

The contrast with *two students* is immediate. In that NP, the determiner *two* fills $\text{Det}(n, d)$, while the nominal *students* fills $\text{Head}(n, h)$. That NP satisfies $\text{CanonicalDetHeadNP}(n)$, not $\text{FusedDetHeadNP}(n)$.

The genitive case shows why the analysis cannot be stated as a lexical category predicate. In *Kim's proposal was stronger than Pat's*, the fused daughter in the second comparison is a genitive NP. It still fills the Determiner and Head functions together. Any claim about its possessive semantics would be an additional bridge claim, not part of $\text{FusedDetHeadNP}(n)$.

5. Open questions

- (a) The inventory of syntactic determiner types needs checking. DP and genitive NP are included here because they expose the main category/function contrast, but CGEL's full set of determiners may need a richer definition.
- (b) Semantic bridges are deliberately absent. Claims about quantification, comparison, partitivity, or possessive interpretation need separate semantic vocabulary plus bridge conditions.
- (c) Partitives probably require tighter treatment of the *of*-PP: it is not just any complement PP.
- (d) The inventory of non-fused determiners should be kept separate from the fused-function configuration unless the interaction is explicitly modelled.
- (e) CGEL page references and the Payne–Huddleston–Pullum treatment of fused functions need to be checked before this becomes a citable fragment.